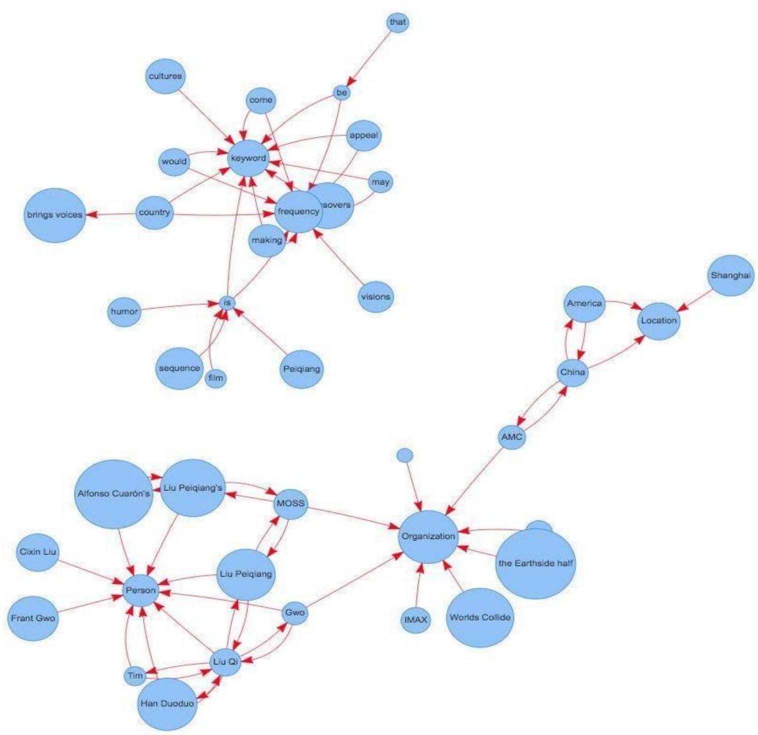
Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Artificial Intelligence (01CT0616)	**Aim:** Representation of a document with their keywords using TextRank	
Experiment No: 09	**Date:**	**Enrolment No:92301733042**

<u>Aim:</u> Representation of a document with their keywords using TextRank

<u>IDE:</u> Google Colab

<u>Theory:</u>

TextRank is an algorithm based on PageRank, which often used in keyword extraction and text summarization. In this article, I will help you understand how TextRank works with a keyword extraction example and show the implementation by Python.

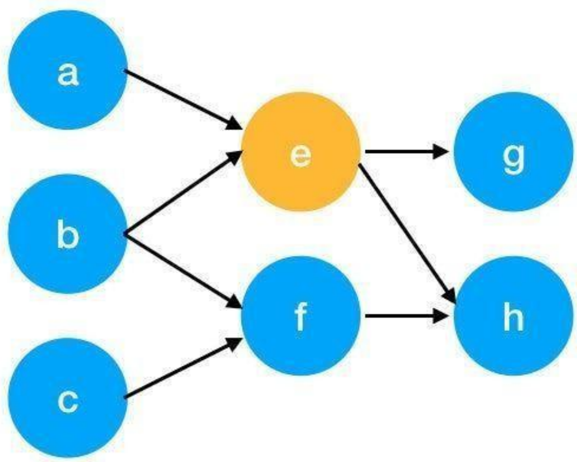


Understand PageRank

There are tons of articles talking about PageRank, so I just give a brief introduction to PageRank. This will help us understand TextRank later because it is based on PageRank. PageRank (PR) is an algorithm used to calculate the weight for web pages. We can take all web pages as a big directed graph. In this graph, a node is a webpage. If webpage A has the link to web page B, it can be represented as a directed edge from A to B. After we construct the whole graph, we can assign weights for web pages by the following formula.

$$S(V_i) = (1-d) + d * \sum_{j \in In(v_i)} \frac{1}{|Out(V_j)|} S(V_j)$$

- S(Vi) - the weight of webpage i
- d - damping factor, in case of no outgoing links
- In(Vi) - inbound links of i, which is a set
- Out(Vj) - outgoing links of j, which is a set
- |Out(Vj)| - the number of outbound links



Here is an example to better understand the notation above. We have a graph to represent how web pages link to each other. Each node represents a webpage, and the arrows represent edges. We want to get the weight of webpage e. We can rewrite the summation part in the above function to a simpler version.

$$In(v_e) = \{a, b\}, j \in \{a, b\}$$

$$\sum_{j \in \{a,b\}} \frac{1}{|Out(V_j)|} S(V_j) = \frac{1}{|Out(V_a)|} S(V_a) + \frac{1}{|Out(V_b)|} S(V_b)$$
$$= \frac{1}{|\{e\}|} S(V_a) + \frac{1}{|\{e, f\}|} S(V_b)$$
$$= S(V_a) + \frac{1}{2} S(V_b)$$

We can get the weight of webpage e by the following function.

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$$S(V_e) = (1-d) + d * \left(S(V_a) + \frac{1}{2} S(V_b) \right)$$

We can see the weight of the webpage e is dependent on the weights of its inbound pages. We need to run this iteration much time to get the final weight. In the initialization, the importance of each webpage is 1.

Pre Lab Exercise:

1. What are the key components of TextRank?
   - **Graph Construction**
     Words or sentences are represented as nodes, and edges show relationships (similarity or co-occurrence).
   - **Edge Weighting**
     Assigns importance based on how strongly nodes are related.
   - **Ranking Algorithm (PageRank)**
     Iteratively calculates importance scores of nodes.
   - **Convergence**
     Repeats calculation until scores stabilize.
   - **Selection**
     Top-ranked words → keywords
     Top-ranked sentences → summary
2. What are some limitations of TextRank for keyword extraction?
   - Does not understand **context or meaning**
   - May select **frequent but less meaningful words**
   - Ignores **word semantics and synonyms**
   - Performance depends on **text quality and length**
   - Cannot handle **domain-specific importance properly**
3. What are the advantages of TextRank for keyword extraction?
   - U**nsupervised** (no training data required)
   - Simple and easy to implement
   - Works well for **general text**
   - Language-independent (with preprocessing)
   - Captures **important words based on relationships**

Program (Code):

```
import networkx as nx
import string
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize
import nltk

# downloading the packages
nltk.download('stopwords')
nltk.download('punkt')
nltk.download('punkt_tab') # Added to address the LookupError

def textrank_keywords(text, topn=5, window_size=4):
    words = word_tokenize(text.lower())
    stop_words = set(stopwords.words('english'))
    words = [w for w in words if w not in stop_words and w not in string.punctuation]

    graph = nx.Graph()

    for i, word in enumerate(words):
        if word not in graph:
            graph.add_node(word)
        for j in range(i + 1, min(i + 1 + window_size, len(words))):
            if words[j] not in graph:
                graph.add_node(words[j])
            graph.add_edge(word, words[j])

    scores = nx.pagerank(graph)
    ranked_words = sorted(scores.items(), key=lambda x: x[1], reverse=True)
    keywords = [word for word, score in ranked_words[:topn]]
    return keywords

text = """Hello my name in Ansh Raythatha,currently i am studing at Marwadi University located
at , Rajkot, Gujarat,lectures timings are 8:00 AM to 1:20 PM , from Monday to Friday """
keywords = textrank_keywords(text, topn=8)
print(keywords)
```

Results:

```
[nltk_data]   Unzipping tokenizers/punkt_tab.zip.
['8:00', 'currently', 'studing', 'timings', 'lectures', 'marwadi', 'university', 'gujarat']
```

Observation:

TextRank is a graph-based ranking algorithm derived from PageRank and is used for keyword extraction and text summarization. From the theory, it is observed that TextRank represents text as a graph where nodes are words or sentences, and edges represent relationships such as similarity or co-occurrence. The importance of each node depends on its connections with other important nodes. By iteratively updating weights, the algorithm identifies the most significant words or sentences. Overall, TextRank is effective because it does not require training data and works based on the structure of the text.

<table>
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Post Lab Exercise:

1. Write about "your ownself" (in not more than 500 words→ You know better about you, rather than ChatGPT!!). Extract top-25 keywords for your portfolio and analyze it in 3 categories: keyword can not give the real idea, keyword gives the real idea, keyword is irrelevant. Paste the code, your portfolio, output and your analysis.

   CODE:

```
import spacy
# Load model
nlp = spacy.load("en_core_web_sm")
# Add TextRank pipeline
nlp.add_pipe("textrank")
text = """I am Ansh Raythatha, a third-year B.Tech student at Marwadi University, pursuing Information and
    Communication Technology.
I have a strong interest in software development and artificial intelligence.
I have worked on projects such as an e-commerce website, a bank management system, and a logistics project
    (PackPal) involving route optimization and live tracking.
I also have experience with microcontrollers and face detection systems using hardware integration."""
doc = nlp(text)
print("Top Keywords:\n")
for phrase in doc._.phrases[:25]:
  print(phrase.text)
```

   OUTPUT:

```
student
detection
marwadi
face
microcontrollers
university
pursuing
experience
also
information
communication
tracking
technology
live
strong
optimization
interest
route
software
involving
packpal
development
project
artificial
systems
```

1. Keywords that Give Real Idea

- software development
- artificial intelligence
- route optimization
- hardware integration
- microcontrollers
- detection systems
- an e-commerce website
- a bank management system
- a logistics project

- PackPal

2. Keywords that Cannot Give Full Idea

- projects
- experience
- tracking
- Information and Communication Technology
- B.Tech
- a third-year B.Tech student
- third-year
- a strong interest

3. Irrelevant Keywords

- Ansh Raythatha (just name)
- Marwadi University (institution only)
- I

Analysis

- TextRank successfully extracted:
  - Important technical keywords
  - Key projects and domains
- However, it also included:
  - General words (projects, experience)